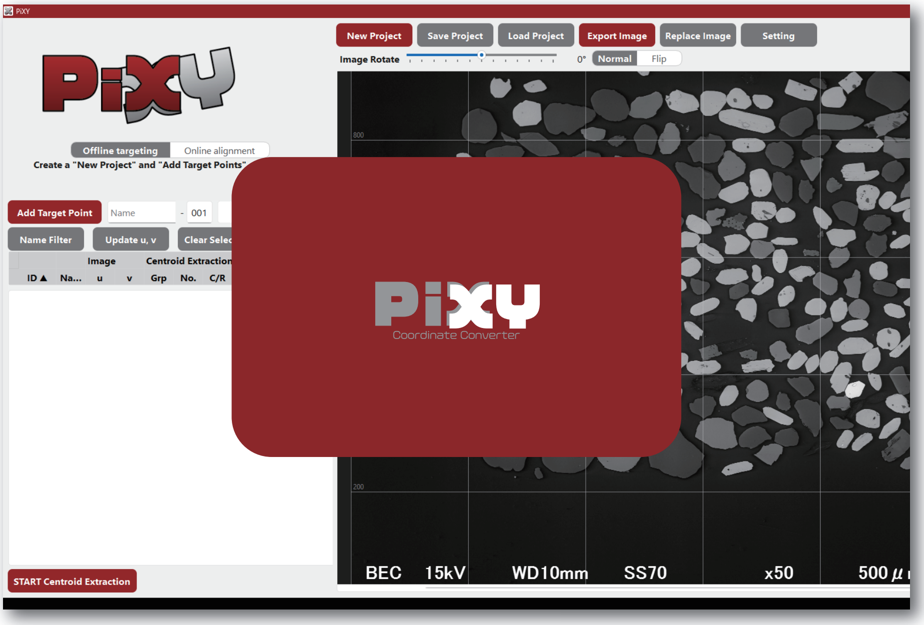
Quick Manual for
Offline Targeting

Developed by Y. Kon (Geological Survey of Japan, AIST)
Version 1.5.1 (2026/8/17)

1

Launch PiXY.exe at office to Start "Offline Targeting"

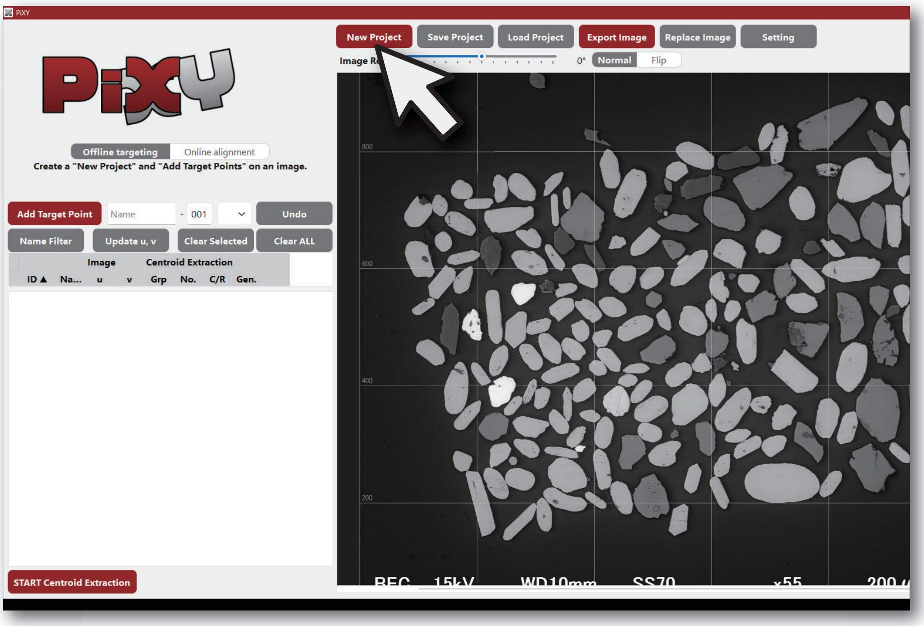
If PiXY.exe is blocked by Windows security settings, try the distributed folder version, modify the Windows security settings, or launch PiXY from Python.



2

Start `New Project` and load a sample image

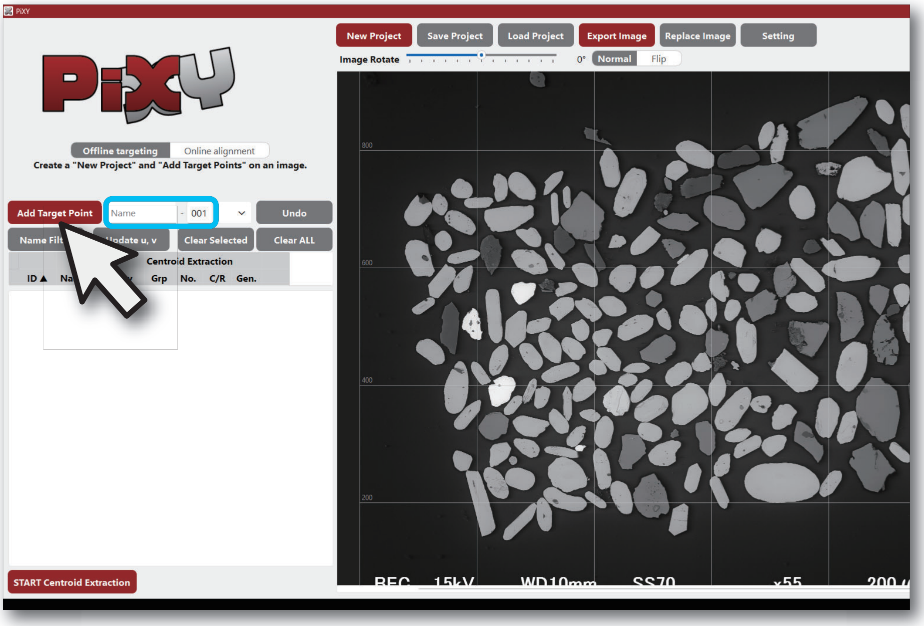
PiXY supports .png, .jpg (.jpeg), .bmp, and .tif (.tiff) files.



3

Click `Add Target Point` to start targeting

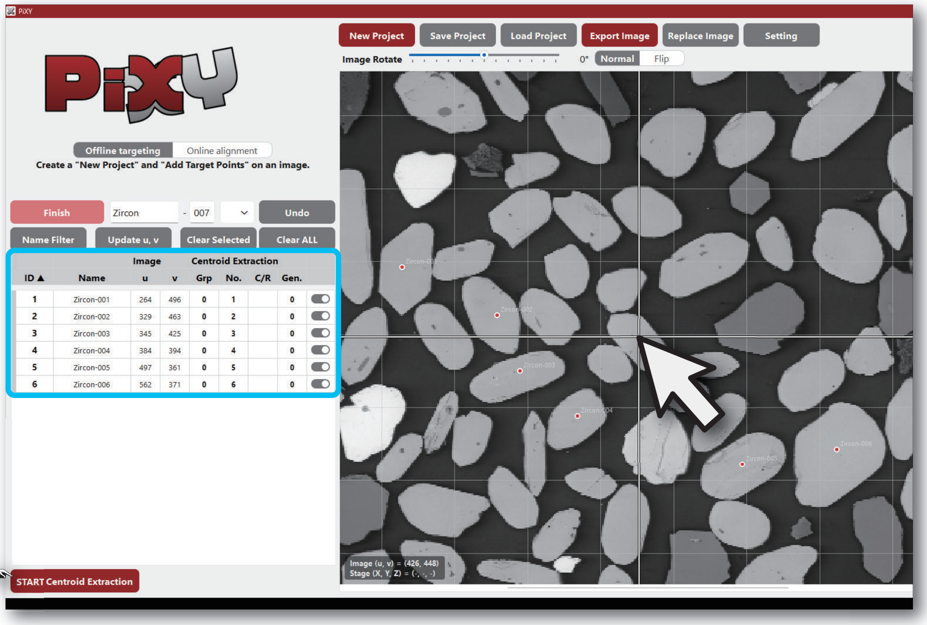
Each new Target Point is assigned the name entered in the "Name" field, followed by an automatically incremented number.



4

Click the sample image to add a Target Point to the Target List

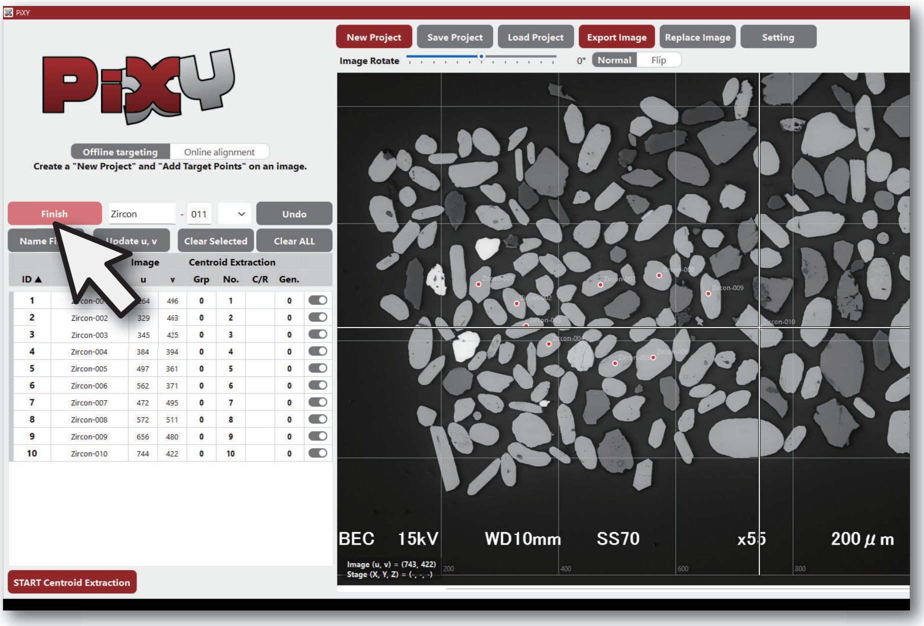
For particle samples, you can also try automatic `Centroid Extraction` mode.



5

Click `Finish` to complete Target Point addition

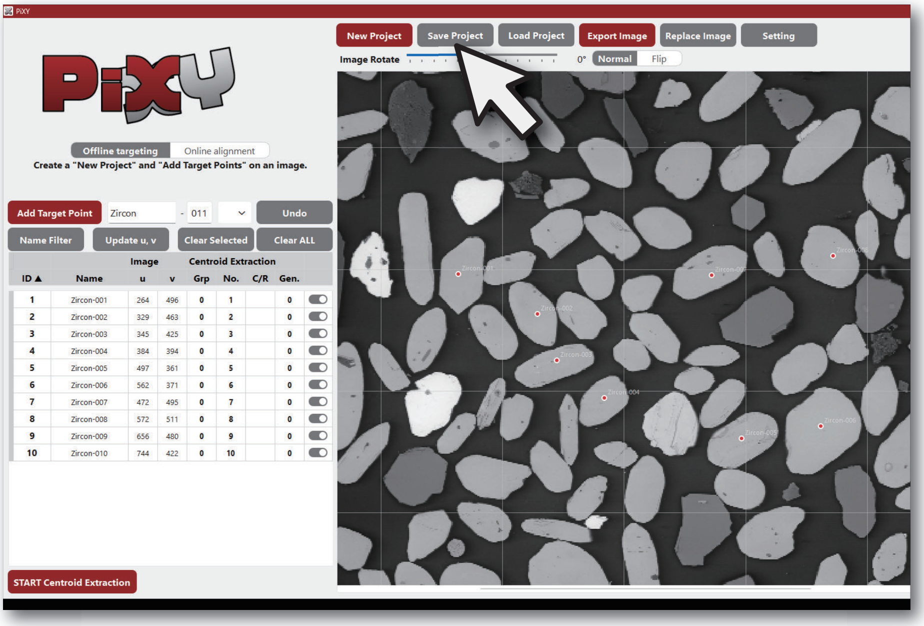
After adding at least one Target Point, the mode automatically ends when the cursor moves outside the sample image.



6

`Save Project` to complete "Offline Targeting"

The loaded image and all entered Target Points are saved in a project file (.pixy).



Centroid Extraction Mode

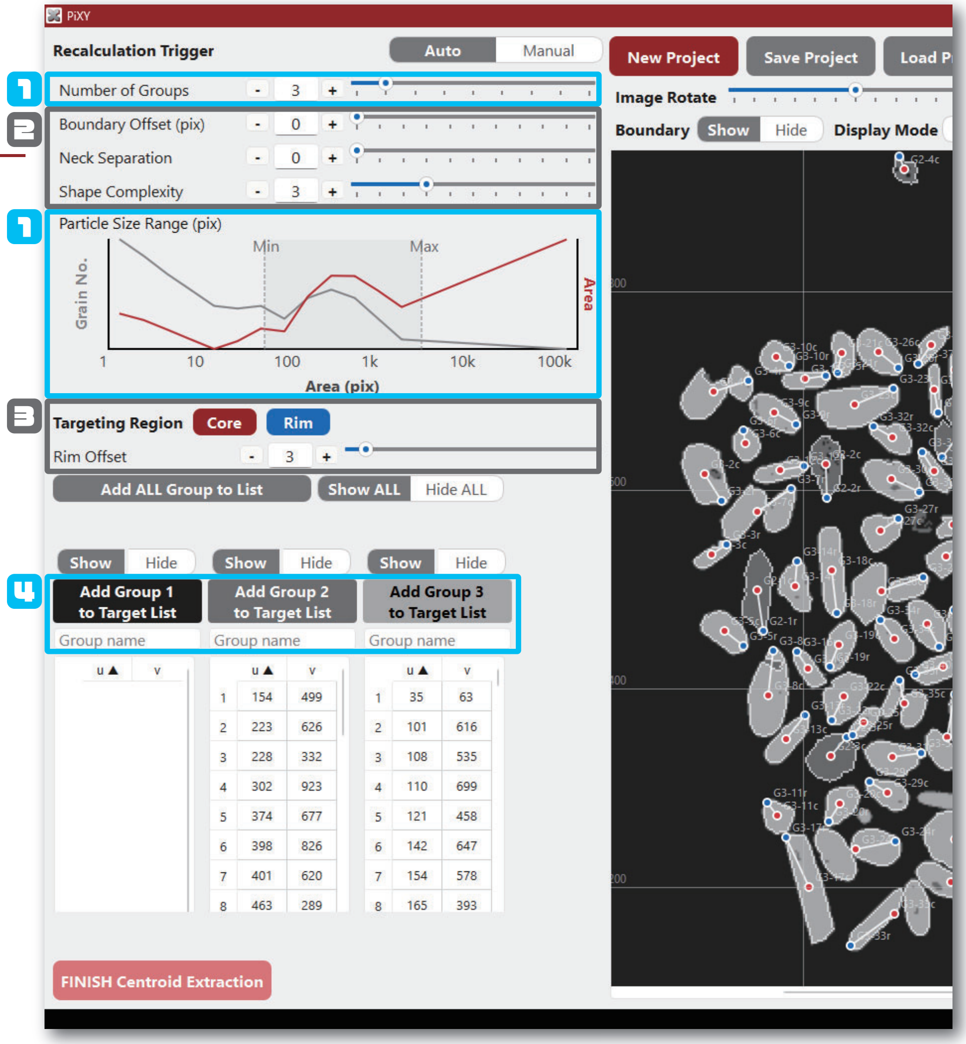
Extract Target Points from the image

1. Set the number of groups for K-means posterization and the particle size range to detect particles.

2. [advanced] *Fine-tune the boundary-, neck-, and shape-related parameters to separate adjacent particles.*

3. [advanced] *Enable core–rim detection and set the target distance from the rim boundary.*

4. Add the desired particle group to the Target List.



Citation

Kon, Y. (2026) "PiXY: Pixel to stage-XY Coordinate Converter". Zenodo. Available at: https://doi.org/10.5281/zenodo.18174474.

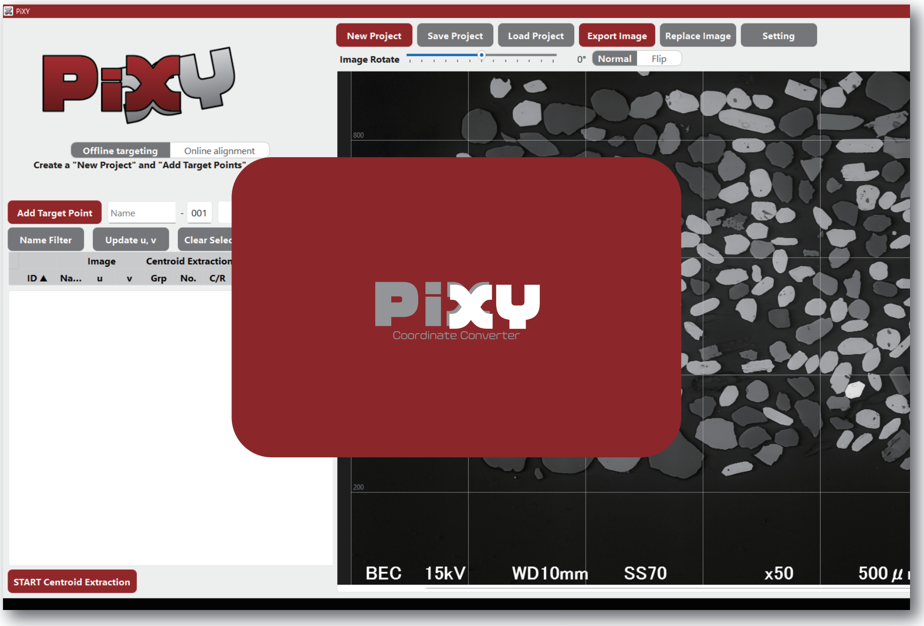
PiXY

Quick Manual for
Online Alignment

Developed by Y. Kon (Geological Survey of Japan, AIST)
Version 1.5.1 (2026/8/17)

1

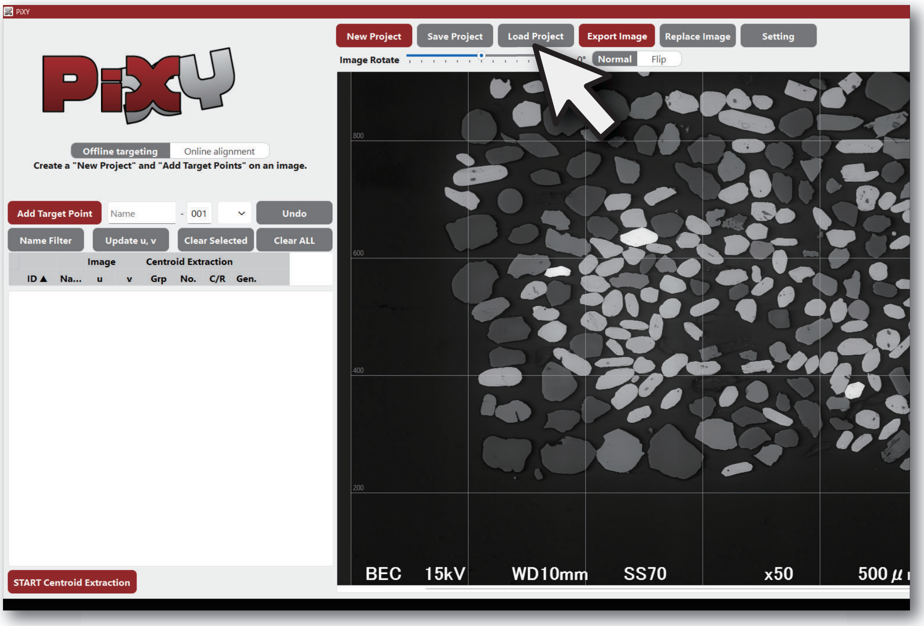
Launch PiXY.exe at laboratory to Start "Online Alignment"



Coordinate transfer is easier when PiXY is run on the instrument control PC. When using your own PC in the laboratory, the exported coordinate file must be copied to the instrument control PC.

2

Click `Load Project` and select a project file (.pixy)

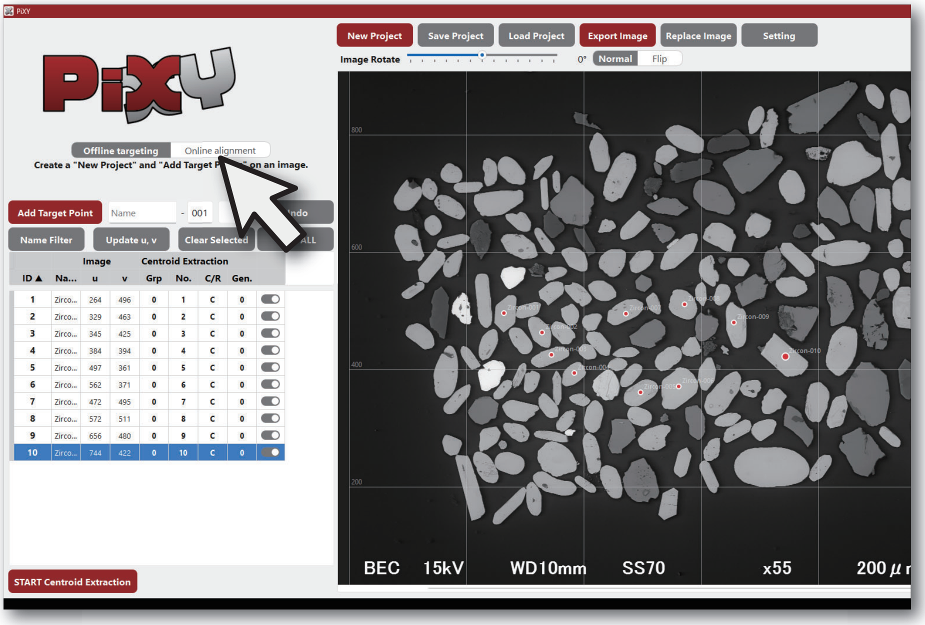


If PiXY is already running in Offline Targeting mode and the same PC is brought into the laboratory, skip these steps and proceed to Step 3.

3

Click “Online Alignment”

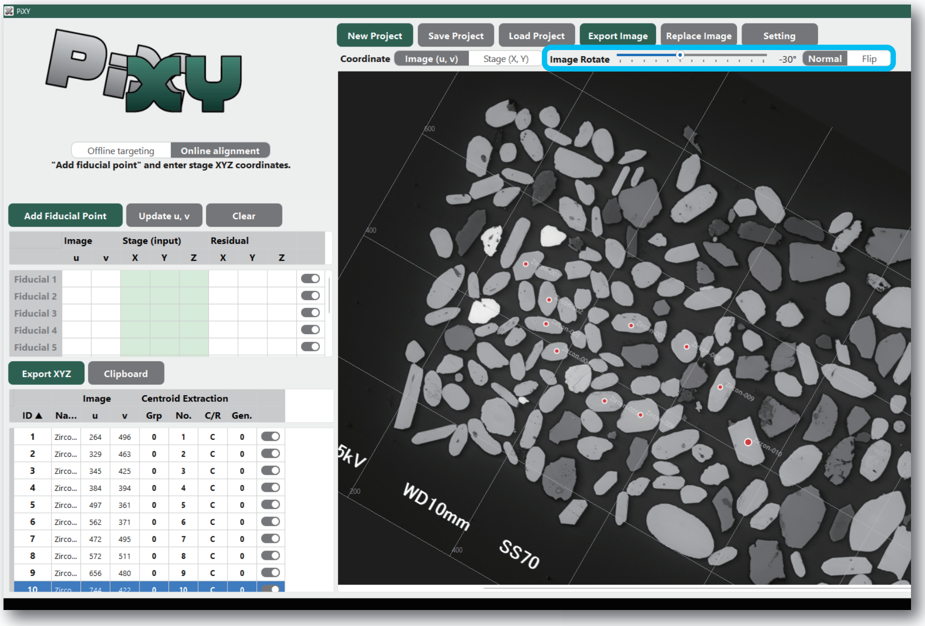
The UI then switches to Online Targeting mode.



4

Click `Add Fiducial Point`

Rotate or flip the sample image to match the sample orientation on the instrument, making it easier to locate the Fiducial Points and determine their stage coordinates.



5

Select a Fiducial Point on the image and enter its XYZ coords on the instrument

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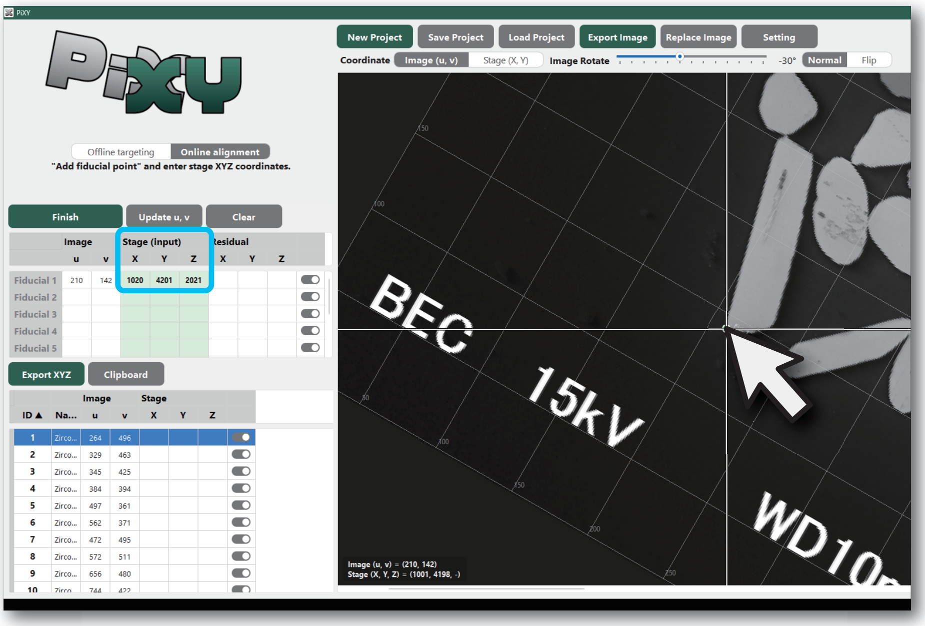
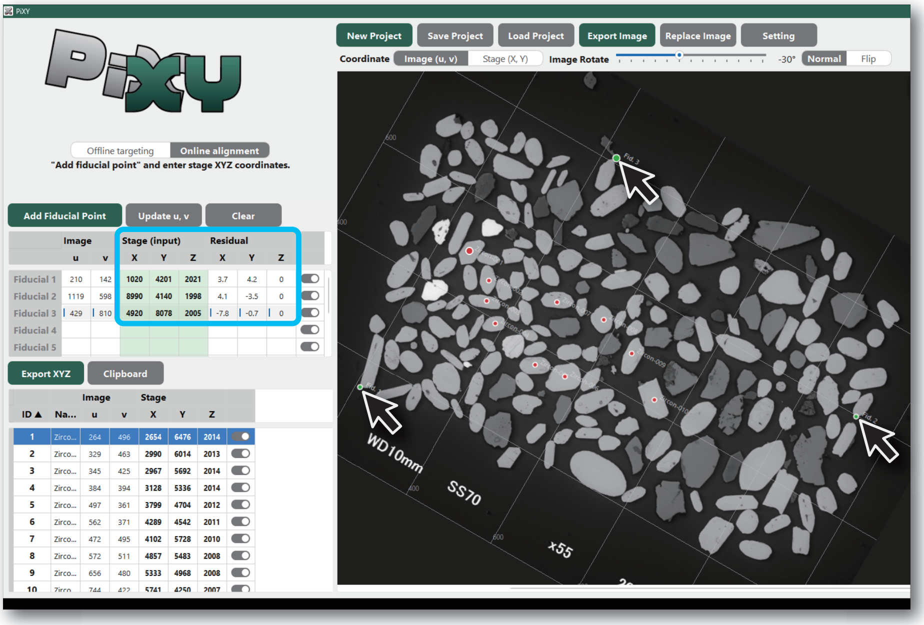


6

Continue adding Fiducial Points

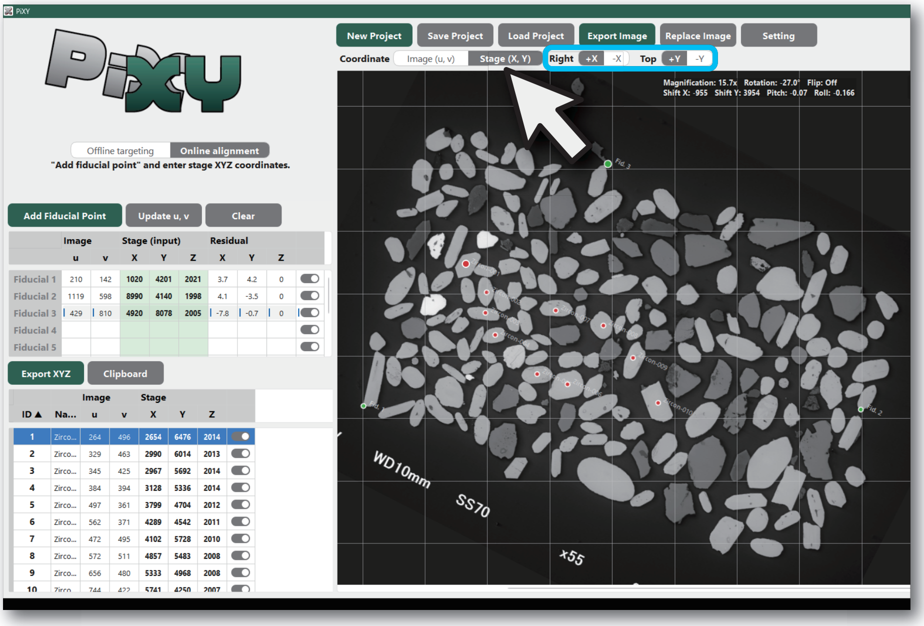
Residuals are calculated for each Fiducial Point, allowing you to check for input errors.



7

The image and coordinate axes are reoriented to match the selected stage coordinate system

Select the X and Y stage directions corresponding to the right and top of the screen.



8

Export the calculated XYZ stage coordinates

Export the coordinates as a CSV file or copy them to the clipboard.

Load them into the instrument software to move the stage near the predefined Target Points.